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(72) Inventors; and

(71) Applicants: **KAYA, Ali** [TR/TR]; Barbaros hayrettin pasha area, 1988 st, NO 4, papatya1 residence, floor 3, unit 36, Esenyurt, 34522 Istanbul (TR). **OJAGHLOO, Hamid** [IR/IR]; mahmutbey mah,dilmenler cad, core living sitesi, b blok daire 67, bağcılar, Istanbul 34218 (IR). **KHALILI, Amirhossein** [IR/IR]; 1st floor,1st Farmani building, 8a alley, Niyayesh boulevard, chamran boulevard, Shiraz 7194917646 (IR). **GHANOUN, Sepideh** [IR/IR]; 1st floor,1st Farmani building, 8a alley, Niyayesh boulevard, chamran boulevard, Shiraz 7194917646 (IR).

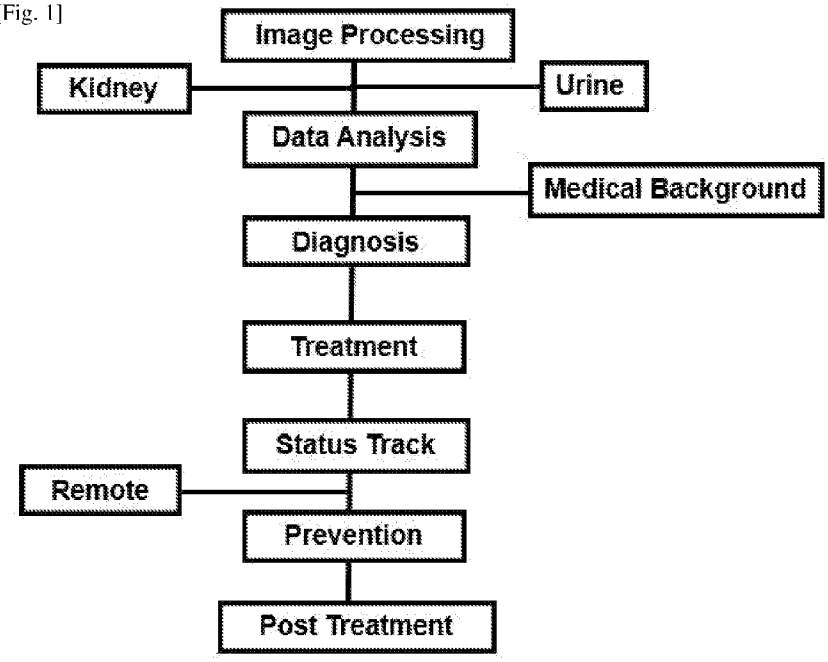
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(54) **Title:** THE AI-POWERED AND REAL-TIME KIDNEY STONE MANAGEMENT SYSTEM

[Fig. 1]




(57) **Abstract:** The smart kidney stone management device utilizes advanced technology, particularly AI, to optimize diagnosis and treatment. Its AI capabilities allow for precise imaging analysis, detecting kidney stones with high accuracy while offering personalized treatment plans based on the patient's unique medical history. The real-time monitoring feature enables urologists to adjust treatments dynamically, ensuring effective care and reducing the likelihood of stone recurrence or new stone formation. The device excels in post-treatment care by tracking recovery and preventing complications, offering remote monitoring and telemedicine consultations for consistent care from a distance. Its data analytics capabilities allow healthcare providers to analyze long-term patient data to enhance treatment strategies and preventive measures.

*[Continued on next page]*

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# Description

## Title of Invention: The AI-powered and Real-time Kidney Stone Management System

[0001] The smart kidney stone management device offers a revolutionary solution for diagnosing, treating, and managing kidney stones. Using advanced imaging systems and AI-powered algorithms, the device captures highly detailed images of the kidneys and urinary tract, precisely identifying stone type, size, and location. It then analyzes patient medical history to recommend the most appropriate treatment, offering personalized care to reduce recurrence and complications.

[0002] The device monitors the patient's progress during and after treatment, allowing real-time adjustments to ensure optimal outcomes. This improves care for both urologists and patients by streamlining diagnostics, providing tailored treatment plans, and enabling remote monitoring. Its data analytics capabilities also enhance long-term patient care, making it a breakthrough in managing kidney stones.

### Technical Field

[0003] A61B - A61B 5/20 - A61P 13/12 - G06T 7/00 - G06F 19/00

### Background Art

[0004] US20230390046

[0005] KIDNEY STONE MANAGEMENT

[0006] Devices, systems, and methods for kidney stone management are provided. A device can include a medical apparatus comprising a kidney stone blocker configured to (i) remain in a human kidney after deployment and (ii) prevent passage of a kidney stone greater than a predetermined size into a ureter while allowing urine to pass to the ureter, and an elongated, tubular member mechanically coupled to the kidney stone blocker and configured to be situated in the ureter.

[0007] The mentioned patent is an apparatus and method for managing and taking care of kidney stones but it is not performed by an application, and also it is not remote while our mentioned patent is useful for patients to monitor their progress through online platforms and get real-time suggestions.

[0008] US10825567B1

[0009] Apparatus and method for informed personal well-being decision making

[0010] An apparatus and method for informed personal-well-being decision making that provides a user with alerts and information, focused on health and wellness, on items they choose for possible consumption. Some embodiments include optical, sonic, smell and other sensors, communications with databases that identify ingredients and effects on health and well-being, as well as user inputs. From user input, GPS,

local conditions and alerts, some embodiments determine information specific to the user and their environment. By using established, and creating new, databases, some embodiments compile, compare, transmit and store data on various consumables. Some embodiments provide access to information on the companies, manufacturers, and various other components in an item's trip from dirt to table. Some embodiments establish methods and procedures to ascertain both the point-of-origin and where the consumable has traveled. Some embodiments provide a score for the specified consumable to show the quality of health provided by the consumable.

[0011] The said invention is close to our claimed patent because both of them offer methods to monitor well-being but a huge difference between them is that our claimed application is exclusively designed for monitoring kidney stones.

[0012] US20220130554A1

[0013] Method for an ai powered automated analyzer for displaying comprehensive medical reports

[0014] Method for an AI Powered Automated Analyzer for Displaying Comprehensive Medical Reports, a Supervised Learning ML classifier for Automated Protocol Change, a Reinforcement Learning ML classifier for Predetermined Protocol Change, an iterative or recurrent neural network natural language processor and a successive or recursive neural network natural language processor for medical learning, understanding, prediction, and prescription.

[0015] The said invention and our claimed one have common purposes for an AI-powered medical system but our invention is designed to monitor and diagnose kidney stones and it contains an image processor to analyze kidney stone related symptoms.

[0016] United States Patent Application 20140242609

[0017] Diagnosis and treatment of kidney stones, methods and compositions therefor

[0018] Methods of detecting, diagnosing, monitoring and treating kidney stones are disclosed. In some embodiments, methods of detecting, diagnosing or monitoring kidney stones comprise contacting a urine sample with anti-Claudin-14 antibody, and detecting quantity of a complex comprising Claudin-14 and the antibody, wherein an increase compared to control levels is diagnostic for kidney stones. In some embodiments, methods further comprise testing a second sample at a second time point to detect increased kidney stones. In some embodiments, methods of treating kidney stone disease comprise administering an miR-9 mimic or an miR-374 mimic. In some embodiments, methods comprise administering an inhibitor of CaSR signaling. In some embodiments, methods comprise administering a HDAC inhibitor. In some embodiments methods of treating hyperparathyroidism and hypercalcemia are disclosed, comprising administering an agonist of CaSR. Methods of abrogating

CaSR-mediated regulation of claudin-14, microRNAs and urinary Ca++ excretion are disclosed, comprising administering a calcineurin inhibitor.

[0019] The mentioned method is for diagnosing kidney stones and offering its antibody, it does the same as our claimed method but their way of diagnosing is different because our system functions remotely and the patients can use it to take cere of their procedure in real-time treatments that our system presents.

[0020] CN116523940

[0021] Contour analysis system for kidney stone image

[0022] The invention relates to the field of medical information processing, and particularly discloses a kidney stone image contour analysis system which comprises an image acquisition unit, an image preprocessing unit, a kidney stone segmentation unit, a puncture needle path analysis unit and a puncture needle path output unit. According to the method, a kidney stone three-dimensional digital image is constructed, slicing analysis is performed on the kidney stone image, a kidney stone contour point set is constructed, conjoint analysis is performed by using points in the kidney stone contour point set and the mass center of the kidney stone, and optimization of an optimal operation path needle insertion point and a needle insertion angle is realized by using an optimal path optimization algorithm. The kidney stone image is analyzed according to the medical image, and the contour, peripheral artery and vein blood vessels of the kidney stone are segmented and recognized.

[0023] The said system is one of the most important aspects of our claimed system which is image processing but in our patent, it is done remotely and in real-time by the users and patients themselves with the help of artificial intelligence while the mentioned patent is a whole process of kidney stone image processing and analysis with another method.

[0024] CN106250714

[0025] Kidney stone prediction method and prediction system based on incremental neural network model

[0026] The invention discloses a kidney stone prediction method based on an incremental neural network model. The kidney stone prediction method comprises the following steps: establishing a kidney stone daily data database; training the neural network model; acquiring daily life data and sending to a server, and storing the daily life data into a user daily data record sheet; extracting data of the current day from the user daily data record sheet to form an n-dimensional vector, normalizing the n-dimensional vector, and then inputting the n-dimensional vector into a kidney stone pathology neural network model for kidney stone probability prediction; judging, by intelligent domestic kidney stone nursing equipment, whether a kidney stone probability value is greater than 0.5, wherein if a user is judged to get kidney stone,

the user goes to a hospital for checking, a checking result is sent back to the server through the intelligent domestic kidney stone nursing equipment, and the server judges whether the checking result is correct; and if the checking result is wrong, performing an incremental algorithm, and dynamically correcting the neural network model. The kidney stone prediction method is accurate in prediction, and the neural network model is customized for each user

[0027] The mentioned method is also one of the initial features of our claimed patent which is prediction and it performs it by monitoring the patients' daily life, but as it is said, our claimed program diagnoses the kidney stone and offers its treatment too.

## Summary of Invention

[0028] A device for remote diagnosis, treatment, and management of kidney stones uses AI algorithms and real-time tracking. Detailed images of the kidney and urinary tract are captured by an imaging system and sent to a central processor for analysis. The processor uses AI to diagnose the stone's features and personalizes treatment based on the patient's medical history. Daily treatment progress is monitored through a mobile phone or online platform, with physicians providing real-time suggestions. The device also tracks dietary, exercise, and genetic patterns to prevent new stone formation and recommends hydration levels for patients. Post-treatment Follow-up Care includes monitoring recovery progress using smart sensors and mobile apps to prevent potential issues.

## Technical Problem

[0029] Kidney stones are a common and painful urological condition affecting millions globally, with a high recurrence rate of nearly 50% within 10 years. Traditional treatments like shockwave lithotripsy and ureteroscopy, while effective, have limitations such as incomplete stone removal and time-consuming diagnostics.
The need for precise diagnosis and monitoring is critical for improving outcomes. Advances in AI, smart devices, and sophisticated imaging systems are revolutionizing healthcare, offering personalized, real-time monitoring and treatment for kidney stones. These innovations enhance precision, reduce recurrence, and improve patient outcomes compared to traditional methods.

## Solution of Problem

[0030] Smart Imaging System:

[0031] The device simultaneously captures images of various parts of the kidney and urinary tract using advanced imaging technologies (such as ultrasound or high-resolution CT scans) and analyzes the images with artificial intelligence to detect the type, size, and location of the stones.

[0032] This advanced technology allows for more accurate and efficient diagnosis of kidney stones, leading to better treatment options and outcomes for patients.

[0033] Additionally, the use of artificial intelligence helps reduce the risk of human error in interpreting imaging results.

[0034] By combining imaging technologies with AI analysis, healthcare providers can make more informed decisions about the best course of treatment for each patient. This cutting-edge approach ultimately improves patient care and overall outcomes in the management of kidney stones.

[0035] Overall, the integration of artificial intelligence in imaging technology has revolutionized the diagnosis and treatment of kidney stones.

[0036] Healthcare providers can now rely on more precise and reliable information to tailor treatment plans to each patient's unique needs, resulting in improved outcomes and patient satisfaction.

[0037] By leveraging AI technology, healthcare providers can also streamline the process of diagnosing and treating kidney stones, leading to quicker and more efficient care for patients. Additionally, the use of AI in imaging technology has the potential to reduce healthcare costs by minimizing unnecessary procedures and optimizing resource allocation.

[0038] Overall, the integration of AI in the diagnosis and treatment of kidney stones has the potential to revolutionize patient care by enhancing accuracy and efficiency.

[0039] This innovative approach holds promise for not only improving outcomes but also reducing the financial burden on healthcare systems.

[0040] Data Analysis and Treatment Prediction:

[0041] By analyzing the data obtained from the images and the patient's clinical information, the device can suggest the best treatment options (such as lithotripsy, surgery, or medication).

[0042] This information helps physicians make more accurate treatment decisions. Additionally, the device can also provide real-time feedback during procedures, allowing for immediate adjustments to ensure the best possible outcomes for the patient.

[0043] This technology ultimately enhances the overall quality of care and patient satisfaction. Furthermore, the device can track the progress of the treatment plan and provide insights on the effectiveness of different interventions.

[0044] This allows for personalized and optimized care for each patient, leading to better outcomes and faster recovery times.

[0045] By utilizing this advanced technology, healthcare providers can tailor treatment plans to individual patients, improving overall health outcomes.

[0046] The ability to monitor progress in real-time also allows for timely modifications to ensure the most effective and efficient care.

[0047] Ultimately, this results in a more efficient healthcare system that can better meet the needs of patients.

[0048] Additionally, the data collected from these devices can be used to improve future treatment protocols and enhance overall patient care.

[0049] Overall, the integration of advanced technology in healthcare not only benefits individual patients but also contributes to the advancement of medical practices as a whole.

[0050] This continuous improvement in patient care ultimately leads to a more effective and responsive healthcare system.

[0051] Patient Status Tracking:

[0052] By connecting the device to mobile applications or online platforms, patients can remotely monitor the status of their kidney stones, and physicians can review the progress or determine if treatment changes are needed in real-time.

[0053] This technology allows for more efficient and effective management of kidney stone treatment, ultimately improving patient outcomes and reducing the need for frequent in-person appointments. Additionally, it provides a convenient way for patients to track their symptoms and communicate with their healthcare providers. Overall, this remote monitoring system offers a convenient and proactive approach to managing kidney stone treatment.

[0054] It empowers patients to take control of their health and allows healthcare providers to intervene promptly when necessary. By utilizing this remote monitoring system, patients can receive timely feedback on their progress and make informed decisions about their treatment plans. This technology bridges the gap between patients and healthcare providers, leading to better communication and personalized care for individuals dealing with kidney stones. Additionally, the remote monitoring system can help reduce the number of in-person visits to healthcare facilities, saving time and resources for both patients and providers. This ultimately leads to more efficient and cost-effective management of kidney stone treatment.

[0055] Preventive Management:

[0056] The device can analyze the patient's dietary, exercise, and genetic patterns and provide recommendations to prevent the formation of new stones. This section can include suggestions for dietary changes or lifestyle modifications. Additionally, the device can track hydration levels and remind the patient to drink more water throughout the day, as proper hydration is crucial in preventing kidney stone formation. It may also offer guidance on specific exercises or activities that can help improve overall kidney health and reduce the risk of future stones.

[0057] Post-treatment Follow-up Care:

[0058] After lithotripsy or surgery, the device can use smart sensors and mobile apps to monitor the patient's recovery process and prevent potential complications. This technology allows healthcare providers to remotely track the patient's progress and intervene, if necessary, ultimately improving overall outcomes and reducing the need for in-person follow-up appointments. Additionally, patients can feel more empowered and engaged in their recovery journey with real-time feedback and support. Overall, this innovative approach to post-operative care can lead to better patient satisfaction and faster recovery times. By utilizing these advanced monitoring tools, healthcare providers can also optimize their resources and provide more personalized care to each patient. Ultimately, this technology can help healthcare systems operate more efficiently and effectively, leading to improved patient care across the board. This proactive approach to post-operative care has the potential to revolutionize the way patients are monitored and supported throughout their recovery process.

## Advantage Effects of the Invention

[0059] - High accuracy in diagnosis and treatment

[0060] - Reduces the need for frequent in-person evaluations

[0061] - Decreases healthcare costs by minimizing unnecessary procedures and optimizing resource allocation

[0062] - Assists physicians in making treatment decisions

## Brief Description of Drawings

[0063] [Fig.1] Shows a flowchart of the whole kidney stone management journey.

## Description of Embodiments

[0064] [Fig.1] presents a flowchart from the first step of the device when it begins its monitoring process which is from taking detailed pictures of the kidneys and urinary tract using advanced technologies like Ultrasound or CT scans. The images are then processed by artificial intelligence algorithms to identify features of kidney stones like size, type, and location. This accelerates diagnosis and informs the physician about the stone type and size. Then, the system uses patient data and medical information to suggest personalized treatment plans. After that, the device offers preventive recommendations based on dietary habits, physical activity, and genetic information to avoid new stone formation and track their regular process. After the treatment, the device also monitors the patient's recovery after procedures like lithotripsy or surgery by collecting vital data and sending it to the physician to prevent complications. Finally, the system can remind patients of upcoming treatments or tests automatically and prevent kidney stone formation again.

## Examples

[0065] Smart Imaging System:

[0066] The device continuously captures images of various areas of the kidneys and urinary tract. These images can be taken using advanced technologies like Ultrasound or CT scans.

[0067] The captured images are sent to a central processor, which utilizes artificial intelligence algorithms to identify specific features of the kidney stone, such as its size, shape, type, and exact location.

[0068] AI accelerates the initial diagnosis of kidney stones with higher precision, informing the physician about the type (e.g., calcium, oxalate, phosphate stones) and size of the stone.

[0069] Data Analysis and Treatment Prediction:

[0070] The data obtained from the images and the patient's medical information (including medical history, clinical tests, symptoms, and underlying causes) are fed into the system.

[0071] Through an AI and machine learning model, the device can suggest the most appropriate treatment method for the patient. This personalized approach allows for more accurate and efficient treatment plans, ultimately improving patient outcomes. Additionally, the system can continuously learn and adapt based on new data, ensuring that recommendations stay up-to-date and relevant. By incorporating real-time data and feedback from healthcare professionals, the system can further enhance its accuracy and effectiveness in providing personalized treatment recommendations. This dynamic learning process ensures that patients receive the most current and evidence-based care available. Overall, this innovative system revolutionizes the way healthcare professionals approach patient care by tailoring treatment plans to individual needs and continuously improving its recommendations. Its ability to adapt and evolve based on new information sets a new standard for personalized medicine in the healthcare industry.

[0072] Patient Status Tracking:

[0073] The patient is connected to the smart device and can view imaging results, kidney health status, and treatment progress through a mobile app or an online platform.

[0074] The physician can remotely track changes in the patient's condition in real time and update the treatment plan if necessary.

[0075] If additional interventions such as dietary or medication changes are needed, the app automatically informs the patient.

[0076] Preventive Management:

[0077] The device uses advanced algorithms to analyze the patient's dietary habits, physical activity, and even genetic information.

[0078] Based on this data, it can offer preventive recommendations to avoid new stone formation, such as:

[0079] Dietary changes (e.g., reducing salt intake, increasing water consumption, limiting oxalate-rich foods).

[0080] Suggesting specific exercises to strengthen the urinary system.

[0081] Recommending particular supplements or medications to prevent stone formation.

[0082] Post-treatment Follow-up Care:

[0083] After lithotripsy or surgery, the device continuously monitors the patient's recovery. Smart sensors connected to the patient's body can collect vital data such as blood pressure, temperature, and kidney function.

[0084] These data are sent to the physician, allowing them to monitor the recovery process and prevent complications.

[0085] The device can also automatically remind the patient of upcoming treatments or periodic tests.

## Industrial Applicability

[0086] The monitoring device is valuable for individuals across all industries, as everyone can benefit from a health tracking and management tool. This is especially crucial for daily use in preventing kidney stones. In the healthcare sector, providers and doctors can monitor their patients remotely at any time and from any location, alleviating concerns about time constraints. Additionally, personal nurses can supervise their clients in real-time. In the pharmaceutical industry, this device can help develop tailored medications based on analyzed data, allowing pharmaceutical companies to recommend appropriate products to those in need.

# Claims

[Claim 1] A remote kidney stone Diagnosis, Treatment, and management system based on AI algorithms and real-time image processing.

[Claim 2] According to claim 1, the system comprises an imaging system to capture detailed pictures of the kidney and urinary tract.

[Claim 3] According to claim 2, the images will sent to the central processor and AI algorithms will analyze and diagnose the kidney stone and its features such as size, type, and location.

[Claim 4] According to claim 3, the central processor also examines the medical history of the patient to give personalized treatment.

[Claim 5] According to claim 4, the device will track the daily procedure of the treatment optically through a mobile phone or online platform.

[Claim 6] According to claim 5, the physicians will review patients' status and give real-time suggestions if needed.

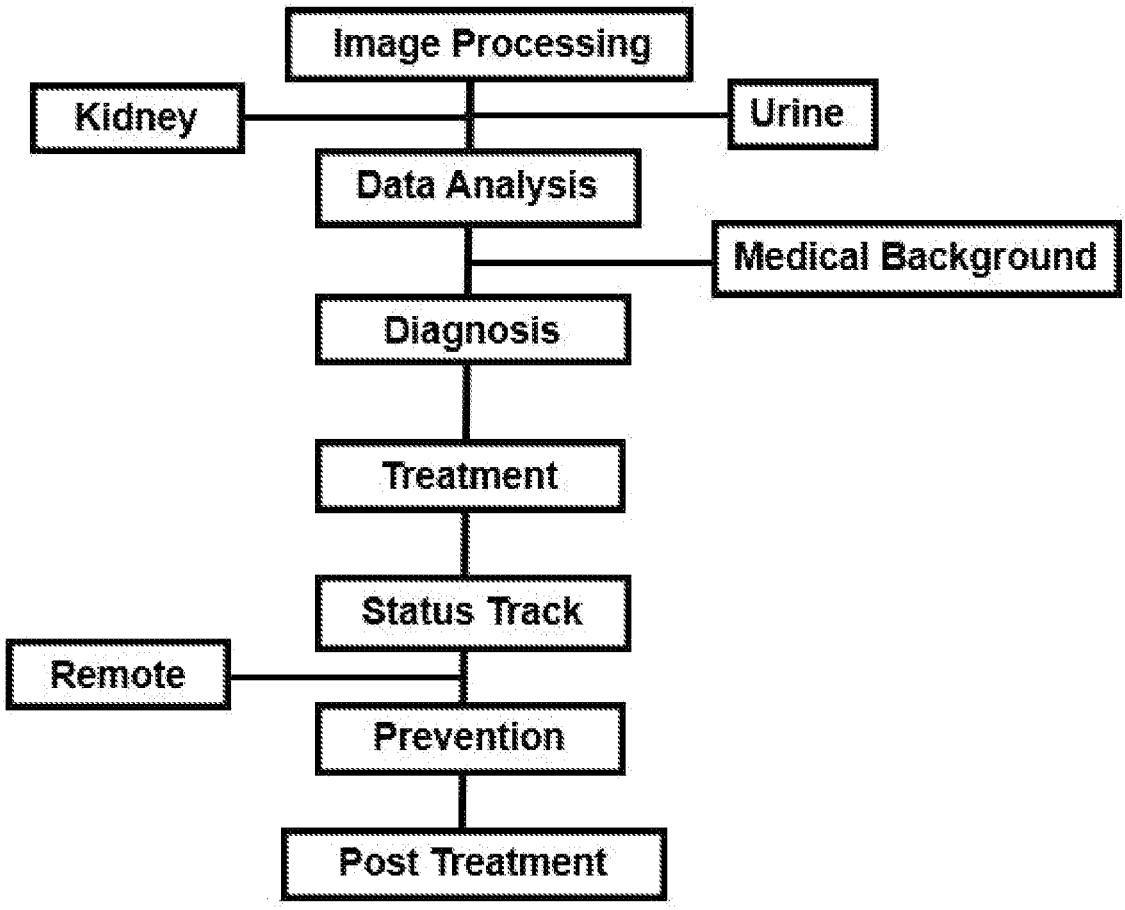
[Claim 7] According to claim 6, The device also investigates the patient's dietary, exercise, and genetic patterns to prevent the formation of new stones.

[Claim 8] According to claim 7, the tracking system checks the hydration level of the patients and recommends a decent amount of water for regular consumption.

[Claim 9] According to claim 1, The device has post-treatment Follow-up Care, especially after lithotripsy or surgery to alert upcoming tests.

[Claim 10] According to claim 9, by using smart sensors and mobile apps to monitor the patient's recovery process, the health providers will take care of the patient's recovery progress and avoid possible problems.

[Fig. 1]


Image Processing
Kidney
Urine
Data Analysis
Medical Background
Diagnosis
Treatment
Status Track
Remote
Prevention
Post Treatment

# INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2024/062820

A. CLASSIFICATION OF SUBJECT MATTER
A61B5/00,G16H50/20,G16H40/67,G06T7/00 Version=2025.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61B,G16H,G06T

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

Keywords:-AI-POWERED,KIDNEY STONE, MANAGEMENT
Databases:-PatSeer, Google Patents,IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

| Category* | Citation of document, with indication, where appropriate, of the relevant passages | Relevant to claim No. |
|---|---|---|
| Y | IN202441000109A-(DR M NARESH KUMAR, DR R CHITRAKALA, MR MAHIILMITHRAN NC, MS VISHALINI NC)(26-Jan-2024)(26-01-2024) Claim 1, Figure 1 | 1-10 |
| Y | WO2015193521A1 (CT DE VISIO PER COMPUTADOR CVC, UNIV BARCELONA AUTONOMA)(23-Dec-2015)(23-12-2015) Abstract, figure 1 | 1-10 |

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

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